THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 48 cm	$AM = \frac{30\times40}{50} = 24$ cm; $AB = 2AM = 48$ cm
2	**(a)** 14 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 7 = 14$ cm
3	**(b)** 49 cm	$15x + 4 = 8x + 25$ gives $x = 3$; $QA = 15(3) + 4 = 49$ cm
4	**(b)** 120 cm	Half-chord $= \sqrt{68^2 - 32^2} = 60$ cm; Chord $= 2 \times 60 = 120$ cm
5	**(c)** 15 cm	$AF = AB - r = 45 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(45 - r) + (60 - r) = 75$ gives $r = 15$ cm
6	**(c)** 36 cm	$PT^2 = OP^2 - OT^2 = 85^2 - 77^2 = 1296$; $PT = 36$ cm
7	**(c)** the perpendicular bisector of AB	$PA = PB$ and $OA = OB$
8	**(d)** 26.88 cm	$AM = \frac{14\times48}{50} = 13.44$ cm; $AB = 2AM = 26.88$ cm
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\angle OPT = 90^\circ$, so $OT^2 = 5^2 + 12^2 = 169$ and $OT = 13$ cm; R is what makes the triangle right-angled.
10	**(c)** A is true but R is false.	A is true; a tangent meets the circle at exactly one point, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of $\triangle OAP = \frac{1}{2} \times 60 \times 45$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{60\times45}{75} = 36$ cm (1); $AB = 2AM = 72$ cm (1).	1+1
12	$CA = CE$ and $DB = DE$ (tangents from C and D); Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$ (1); Perimeter $= 2 \times 9 = 18$ cm (1).	1+1
13	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord; Half-chord $= \sqrt{90^2 - 72^2} = 54$ cm (1); Chord $= 2 \times 54 = 108$ cm (1).	1+1
14	Area of $\triangle OAP = \frac{1}{2} \times 72 \times 21$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{72\times21}{75} = 20.16$ cm (1); $AB = 2AM = 40.32$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of $\triangle OAP = \frac{1}{2} \times 30 \times 40$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{30\times40}{50} = 24$ cm (1); $AB = 2AM = 48$ cm (1).	1+1+1
16	$AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 36 - r$, $CD = BC - r = 77 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(36 - r) + (77 - r) = 85$ gives $r = 14$ cm (1).	1+1+1
17	Tangents from an external point are equal: $QA = QB$ (1); $21x + 5 = 13x + 29$ gives $x = 3$ (1); $QA = 21(3) + 5 = 68$ cm (1).	1+1+1
	OR: Area of $\triangle OAP = \frac{1}{2} \times 14 \times 48$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{14\times48}{50} = 13.44$ cm (1); $AB = 2AM = 26.88$ cm (1).	1+1+1
18	$AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 40 - r$, $CD = BC - r = 75 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(40 - r) + (75 - r) = 85$ gives $r = 15$ cm (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $18x + 7 = 11x + 35$ gives $x = 4$ (1); $QA = 18(4) + 7 = 79$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 90° (tangent is perpendicular to the radius at the point of contact) (1). (ii) $PA = \sqrt{OP^2 - OA^2} = \sqrt{100 - 36} = 8$ m (1). (iii) $PB = PA = 8$ m, tangents from an external point are equal (1). (iv) $\triangle OAP \cong \triangle OBP$, so area $= 2 \times \frac{1}{2} \times 6 \times 8 = 48$ m^2 (2).	1+1+1+2

20	Let the incircle touch AB, BC, CA at F, D, E and let the centre be O; $OF \perp AB$, $OD \perp BC$ (1); $BFOD$ has three right angles and $OF = OD = r$, so it is a square: $BF = BD = r$ (2); $AE = AF = AB - r$ and $CE = CD = BC - r$ (equal tangents) (1); $AC = AE + EC = AB + BC - 2r$, so $r = \frac{AB+BC-AC}{2}$ (1).	1+2+1+1
	OR: Area of $\triangle OAP = \frac{1}{2} \times 36 \times 27$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (2); $AM = \frac{36 \times 27}{45} = 21.6$ cm (2); $AB = 2AM = 43.2$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 112 cm	Half-chord $= \sqrt{70^2 - 42^2} = 56$ cm; Chord $= 2 \times 56 = 112$ cm
2	**(d)** 12 cm	$AF = AB - r = 30 - r$, $CD = BC - r = 72 - r$ and $AF + CD = AC$ (tangent lengths); $(30 - r) + (72 - r) = 78$ gives $r = 12$ cm
3	**(b)** 40 cm	$PT^2 = OP^2 - OT^2 = 85^2 - 75^2 = 1600$; $PT = 40$ cm
4	**(b)** $\frac{a + b - c}{2}$	$r = s - c$ for a right angle
5	**(d)** 38.4 cm	$AM = \frac{24 \times 32}{40} = 19.2$ cm; $AB = 2AM = 38.4$ cm
6	**(c)** 26 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 13 = 26$ cm
7	**(d)** 54 cm	$17x + 3 = 9x + 27$ gives $x = 3$; $QA = 17(3) + 3 = 54$ cm
8	**(c)** 110 cm	Half-chord $= \sqrt{73^2 - 48^2} = 55$ cm; Chord $= 2 \times 55 = 110$ cm
9	**(d)** A is false but R is true.	From an interior point no tangent can be drawn, so A is false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A is the definition of a tangent and R is also true (one at each point), but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$CA = CE$ and $DB = DE$ (tangents from C and D); Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$ (1); Perimeter $= 2 \times 16 = 32$ cm (1).	1+1
12	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord; Half-chord $= \sqrt{91^2 - 84^2} = 35$ cm (1); Chord $= 2 \times 35 = 70$ cm (1).	1+1
13	Area of $\triangle OAP = \frac{1}{2} \times 36 \times 27$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{36 \times 27}{45} = 21.6$ cm (1); $AB = 2AM = 43.2$ cm (1).	1+1
14	$CA = CE$ and $DB = DE$ (tangents from C and D); Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$ (1); Perimeter $= 2 \times 21 = 42$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of $\triangle OAP = \frac{1}{2} \times 24 \times 32$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{24 \times 32}{40} = 19.2$ cm (1); $AB = 2AM = 38.4$ cm (1).	1+1+1
16	$AC = 85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 51 - r$, $CD = BC - r = 68 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(51 - r) + (68 - r) = 85$ gives $r = 17$ cm (1).	1+1+1
17	Tangents from an external point are equal: $QA = QB$ (1); $16x + 9 = 9x + 30$ gives $x = 3$ (1); $QA = 16(3) + 9 = 57$ cm (1).	1+1+1
	OR: Area of $\triangle OAP = \frac{1}{2} \times 45 \times 60$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{45 \times 60}{75} = 36$ cm (1); $AB = 2AM = 72$ cm (1).	1+1+1
18	$AC = 87$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 60 - r$, $CD = BC - r = 63 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(60 - r) + (63 - r) = 87$ gives $r = 18$ cm (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $23x + 4 = 15x + 20$ gives $x = 2$ (1); $QA = 23(2) + 4 = 50$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 90° (1). (ii) In quadrilateral OAPB, $\angle AOB = 360^\circ - 90^\circ - 90^\circ - 80^\circ = 100^\circ$ (1). (iii) OP bisects $\angle APB$, so $\angle OPA = 40^\circ$ (1). (iv) $OA = OB$, so $\angle OAB = \frac{180^\circ - 100^\circ}{2} = 40^\circ$ (1); $PA = PB$, so $\angle PAB = \frac{180^\circ - 80^\circ}{2} = 50^\circ$ (1).	1+1+1+2
20	Area of $\triangle OAP = \frac{1}{2} \times 21 \times 28$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (2); $AM = \frac{21 \times 28}{35} = 16.8$ cm (2); $AB = 2AM = 33.6$ cm (1).	2+2+1

	OR: $AC = \sqrt{35^2 + 84^2} = 91$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 35 - r$ and $CD = CE = 84 - r$ and $AE + EC = AC$: $(35 - r) + (84 - r) = 91$ (2); $r = 14$ cm, $AF = 21$ cm, $CD = 70$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90°$; the correct sum is $180°$.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 60°	$3\angle APB = 180^\circ$
2	**(a)** 33.6 cm	$AM = \frac{21\times28}{35} = 16.8$ cm; $AB = 2AM = 33.6$ cm
3	**(d)** 38 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 19 = 38$ cm
4	**(d)** 62 cm	$14x + 6 = 8x + 30$ gives $x = 4$; $QA = 14(4) + 6 = 62$ cm
5	**(c)** 140 cm	Half-chord $= \sqrt{74^2 - 24^2} = 70$ cm; Chord $= 2 \times 70 = 140$ cm
6	**(c)** 8 cm	$AF = AB - r = 18 - r$, $CD = BC - r = 80 - r$ and $AF + CD = AC$ (tangent lengths); $(18 - r) + (80 - r) = 82$ gives $r = 8$ cm
7	**(a)** 60 cm	$PT^2 = OP^2 - OT^2 = 87^2 - 63^2 = 3600$; $PT = 60$ cm
8	**(c)** $2r$	$\sin 30^\circ = r/OP$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Tangent length $= \sqrt{5^2 - 3^2} = 4$ cm, found by using the right angle given in R.
10	**(c)** A is true but R is false.	In quadrilateral OAPB, $\angle OAP = \angle OBP = 90^\circ$, so the other two angles sum to 180°: A true; the tangent is perpendicular, not parallel, to the radius, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord; Half-chord $= \sqrt{95^2 - 76^2} = 57$ cm (1); Chord $= 2 \times 57 = 114$ cm (1).	1+1
12	Area of $\triangle OAP = \frac{1}{2} \times 44 \times 33$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{44\times33}{55} = 26.4$ cm (1); $AB = 2AM = 52.8$ cm (1).	1+1
13	$CA = CE$ and $DB = DE$ (tangents from C and D); Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$ (1); Perimeter $= 2 \times 28 = 56$ cm (1).	1+1
14	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord; Half-chord $= \sqrt{97^2 - 72^2} = 65$ cm (1); Chord $= 2 \times 65 = 130$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of $\triangle OAP = \frac{1}{2} \times 21 \times 72$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{21\times72}{75} = 20.16$ cm (1); $AB = 2AM = 40.32$ cm (1).	1+1+1
16	$AC = 89$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 39 - r$, $CD = BC - r = 80 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(39 - r) + (80 - r) = 89$ gives $r = 15$ cm (1).	1+1+1
17	Tangents from an external point are equal: $QA = QB$ (1); $20x + 8 = 12x + 24$ gives $x = 2$ (1); $QA = 20(2) + 8 = 48$ cm (1).	1+1+1
	OR: Area of $\triangle OAP = \frac{1}{2} \times 27 \times 36$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{27\times36}{45} = 21.6$ cm (1); $AB = 2AM = 43.2$ cm (1).	1+1+1
18	$AC = 90$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF = AB - r = 54 - r$, $CD = BC - r = 72 - r$ and $AF + CD = AC$ (tangent lengths) (1); $(54 - r) + (72 - r) = 90$ gives $r = 18$ cm (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $25x + 3 = 17x + 19$ gives $x = 2$ (1); $QA = 25(2) + 3 = 53$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 90°, since AB is a tangent to the smaller circle and OP is the radius through the contact point (1). (ii) $AP = \sqrt{13^2 - 5^2} = 12$ cm (1). (iii) The perpendicular from the centre bisects the chord, so $AB = 2 \times 12 = 24$ cm (1). (iv) $AP = \sqrt{17^2 - 8^2} = 15$ cm, so the chord $= 30$ cm (2).	1+1+1+2

20	Area of $\triangle OAP = \frac{1}{2} \times 45 \times 60$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (2); $AM = \frac{45 \times 60}{75} = 36$ cm (2); $AB = 2AM = 72$ cm (1).	2+2+1
	OR: $AC = \sqrt{65^2 + 72^2} = 97$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 65 - r$ and $CD = CE = 72 - r$ and $AE + EC = AC$: $(65 - r) + (72 - r) = 97$ (2); $r = 20$ cm, $AF = 45$ cm, $CD = 52$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
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- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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